

BloodLink — Project Handoff & AI Context Guide

Purpose: Read this file at the start of any new Cursor / AI session before making changes. This document captures product decisions, architecture, migration state, and how the app should evolve.

Repository: <https://github.com/sridushiva-dev/NuraniBloodDonationTracker>

App name: BloodLink

Last updated: September 2026

1. What This App Is

BloodLink is a **blood donation request management platform** for India. Users can:

- **Request blood** for a patient (hospital/blood bank location, blood group, units, deadline)
- **Donate** when matched by blood group, eligibility, and location/community rules
- **Join communities** (groups like colleges, neighborhoods, organizations) to share and respond to requests
- **Chat in-app** after a donor accepts (no phone numbers exposed)

The product owner is building this for real-world use (Palakkad / Kerala focus initially, with hospitals across 8 cities).

2. Tech Stack

Layer	Technology
Framework	Next.js 16 (App Router) — read <code>node_modules/next/dist/docs/</code> before changing Next APIs; this version has breaking changes vs training data
UI	React 19, Tailwind CSS v4, Apple-inspired design system
Database / Auth	Supabase (Postgres + RLS + Auth)
Hosting	Vercel (recommended)
Icons	lucide-react

Environment variables (`.env.local`)

```
NEXT_PUBLIC_SUPABASE_URL=https://your-project.supabase.co
NEXT_PUBLIC_SUPABASE_ANON_KEY=your-anon-key
SUPABASE_SERVICE_ROLE_KEY=your-service-role-key
NEXT_PUBLIC_APP_URL=http://localhost:3000 # or production URL
```

Key Supabase clients

- `src/lib/supabase/server.ts` — user-scoped (RLS)
- `src/lib/supabase/admin.ts` — service role (broadcasts, notifications, admin actions)
- `src/lib/supabase/client.ts` — browser client

3. Branch & Deployment State

Branch	Status
main	Stable base: Apple UI, hospital picker, facilities, PIN code location
cursor/communities-feature-4b23	Latest work — communities, auto-approve, donation history, RLS fix

Before continuing development: merge `cursor/communities-feature-4b23` into `main` (or work from that branch).

Open PRs (may change): check GitHub for PR #5 / #6 on the communities branch.

4. Database Migrations — RUN IN ORDER

All migrations live in `supabase/migrations/`. Run each in the **Supabase SQL Editor** in numeric order:

#	File	What it does
001	<code>001_initial_schema.sql</code>	Core schema: profiles, donor_profiles, blood_requests, invitations, chat, notifications, RLS, accept_invitation RPC
002	<code>002_fix_signup_trigger.sql</code>	Fix auth signup trigger
003	<code>003_promote_hareesh_admin.sql</code>	Promote hareesh.nurani96@gmail.com to app admin
004	<code>004_facilities_and_custom_location.sql</code>	facilities table (234 hospitals), custom hospital + PIN fields on requests
005	<code>005_communities.sql</code>	Communities, members, invites, request_communities, pins, notify_community_only, auto-approve pending users
006	<code>006_donation_history.sql</code>	not_donated_reason column on donation_confirmations
007	<code>007_fix_rls_recursion.sql</code>	Required — fixes infinite recursion detected in policy for relation "blood_requests" when creating requests

 *If request creation fails with RLS recursion, migration **007** has not been applied.*

5. Product Rules (Confirmed by Owner)

5.1 Users & onboarding

- **Anyone can sign up** — no global admin approval gate
- Onboarding sets status to **active** immediately (not `pending_approval`)
- Community membership is **optional**
- Light **app admin** remains: `hareesh.nurani96@gmail.com` (role `admin` in `profiles`)

5.2 Blood requests

- Requester selects **hospital from list** (230+ facilities across 8 cities) OR **custom hospital name + 6-digit PIN code**
- Donor location still uses **GPS on profile/onboarding**
- **50 km radius** for generic donor matching and generic notifications
- Blood group match required; optional **replacement blood groups**
- **90-day cooldown** after donation (`DONOR_COOLDOWN_DAYS = 90`)
- **First-come unit acceptance** — 1 accepted donor = 1 unit (atomic `accept_invitation` RPC)
- In-app chat opens after donor accepts

5.3 Communities

Rule	Detail
Who can create	Any active user
Visibility	Creator chooses public (open join) or private (invite link / admin add-by-email)
Posting	Requester must be a member to post to a community; can select multiple communities
Generic wall	All requests always appear on the generic request wall , regardless of communities selected
Community response	Any community member donor can see/respond — no distance limit
Notifications	One notification per donor per request (even if posted to multiple communities)
Generic notifications	Only donors within 50 km
Community notifications	All eligible community member donors (any distance)
Donor preference	<code>notify_community_only</code> on <code>donor_profiles</code> — skips generic notifications outside their communities; still sees all requests in Active Requests list
Distance popup	If donor accepts a request >50 km away, show confirmation: "Are you sure? Location is more than 50 km away"
Pinning	Unlimited community pins on home; long-press drag to reorder
Admin	Community creator = default admin; can add/remove admins and members

5.4 Donor donation tracking

- After **accepting** a request, home shows: "**Have you completed the donation to [Patient name]?**"
 - **Yes** → pick date → saved as `last_donation_date` , availability turned off
 - **No** → must enter a **reason in words** (`not_donated_reason`)
- **Donation history** on Profile page (all past confirmations)
- **Welcome message on home:**
 - If donated: You have saved N lives through your donations
 - If zero donations: motivational message (never show "0 lives")

5.5 Unchanged core rules

- Blood group matching, 90-day cooldown, first-come units, in-app chat — all unchanged from original spec

6. Architecture Overview

```
src/
├─ app/
│   ├── (auth)/          login, signup, verify-email
│   ├── (app)/           protected app (requireActiveProfile)
│   │   ├── home/        dashboard, communities feed, lives saved, donation prompts
│   │   ├── communities/ list, create, detail, join-by-code
│   │   ├── donor/       invites + all active requests browse
│   │   ├── requests/    create, detail
│   │   ├── profile/     account, donor settings, donation history
│   │   ├── notifications/
│   │   └─ chat/
│   ├── admin/           light app admin (users, requests, audit)
│   └─ onboarding/       profile completion
├─ components/
│   ├── communities/     selector, pins, admin panel, join/leave
│   ├── donor/           availability, invites, confirmations, history
│   ├── layout/          app-nav, mobile tabs, more sheet
│   └─ ui/               Apple-style primitives (button, card, grouped-list, switch)
├─ lib/
│   ├── actions/         server actions (requests, profile, communities, admin, chat)
│   ├── facilities/       static hospital data per city (8 cities)
│   ├── auth.ts          requireActiveProfile, requireAdmin
│   ├── types.ts
│   └─ utils.ts          haversine, eligibility helpers
└─ supabase/migrations/
```

Auth gating (src/Lib/auth.ts)

- requireActiveProfile() — used in (app)/layout.tsx
- Redirects: unverified email → /verify-email , incomplete → /onboarding , rejected → /rejected
- pending_approval redirect still exists but users are auto-approved on onboarding

Request broadcast (src/Lib/actions/requests.ts → broadcastRequest)

1. Find eligible donors (willing, available, active, blood match, cooldown OK)
2. **Generic match:** within 50 km → create invitation + notify (unless notify_community_only and not in posted community)
3. **Community match:** members of posted communities at any distance → invitation + notify
4. Deduplicate: one invitation and one notification per donor

Service role vs user client

Operation	Client
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User CRUD under RLS	<code>createClient()</code>
Broadcast, notifications, audit, community linking on create	<code>createServiceClient()</code>

7. Database Tables (Key)

Table	Purpose
profiles	User account, location, status, role
donor_profiles	Blood group, last donation, availability, <code>notify_community_only</code>
blood_requests	Patient requests with location (facility or custom PIN)
donor_invitations	Match records with <code>distance_km</code> , <code>accept/reject</code> status
donation_confirmations	Post-donation yes/no + date or reason
communities	Groups (public/private)
community_members	Membership + <code>is_admin</code>
community_invites	Invite codes for private communities
request_communities	Links requests to communities
user_community_pins	Home screen pin order
facilities	234 seeded hospitals/blood banks
notifications	In-app only (no push/email yet)
chat_threads / chat_messages	Mediated chat

Important RPC

- `accept_invitation(p_invitation_id, p_donor_id)` — atomic unit allocation + chat thread creation

RLS helpers (SECURITY DEFINER — avoid removing)

- `is_admin()`
- `is_community_member()`, `is_community_admin()`
- `is_request_owner()`, `is_invited_donor()` — **prevent blood_requests ↔ donor_invitations recursion**

8. Facilities / Location

- **Static data:** `src/lib/facilities/*.ts` (palakkad, coimbatore, kochi, chennai, bangalore, mumbai, pune, delhi)
- **DB seed:** migration 004 populates `facilities` table
- **Custom location:** hospital name + 6-digit PIN → `src/lib/pincode.ts` (India Post API + Nominatim geocoding)
- **Request form:** `src/components/hospital-location-picker.tsx`

9. UI / Navigation

- **Mobile-first** with bottom tab bar: Home | Donor | Alerts | Chat | More
- **Desktop** nav includes Groups (communities), Profile, Admin (if admin)
- Design tokens in `src/app/globals.css` (iOS-inspired colors, shadows, system fonts)
- Reusable: `GroupedSection` , `GroupedRow` , `PageHeader` , `Switch`

10. Known Issues & Fixes Applied

Issue	Fix
Request creation: infinite recursion detected in policy for relation "blood_requests"	Run migration 007
Hospital picker not visible on mobile	Fixed overflow/z-index in request form
Features not visible after deploy	Ensure <code>main</code> is deployed / branch merged to <code>main</code> on Vercel
<code>pending_approval</code> page still exists	Legacy route; new users skip it (auto-approved)

11. What Is NOT Built Yet (Potential Next Steps)

These were discussed or implied but not fully implemented — confirm with product owner before building:

- ☐ Push / email notifications (currently in-app only)
- ☐ Community feed sorting by "unread since last visit" (currently by active request count)
- ☐ Transfer community ownership before creator leaves
- ☐ Admin dashboard updates for community moderation
- ☐ Remove or repurpose `/pending-approval` page and admin user-approval UI (legacy)
- ☐ Merge `cursor/communities-feature-4b23` → `main` and redeploy Vercel
- ☐ `.env.example` file (referenced in README but may be missing from repo)

12. How to Start a New AI Session

Paste this into Cursor when resuming:

```
Read docs/PROJECT_HANDOFF.md in the BloodLink repo before doing anything.
That file has the full product spec, migration order, architecture, and confirmed
owner decisions.
Current branch with latest features: cursor/communities-feature-4b23 (or main if
merged).
All Supabase migrations 001-007 must be applied in order.
App admin: hareesh.nurani96@gmail.com
```

Local dev

```
npm install
npm run dev
# http://localhost:3000
```

Verify build

```
npm run build
```

13. Owner Decisions Log (Communities Q&A — September 2026)

1. Anyone can join, create requests, browse; generic posting/donating for all
 2. Any user can create a community
 3. Public = open join; private = admin add or invite links; users can leave
 4. Must be community member to post; multi-community select; always on generic wall
 5. Community members respond with no distance limit; >50 km accept shows popup
 6. One notification per request; generic notify ≤50 km only; all requests visible in active list
 7. Community-only notification toggle; never hide requests
 8. Light app admin: hareesh.nurani96@gmail.com
 9. Existing blood rules unchanged
 10. Unlimited pinning with reorder (long-press)
 11. Auto-approve; communities not mandatory
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14. File Quick Reference

Task	Start here
Create request	src/app/(app)/requests/new/page.tsx, src/lib/actions/requests.ts
Donor matching	broadcastRequest() in requests.ts
Accept invite	acceptInvitation(), accept_invitation RPC
Communities CRUD	src/lib/actions/communities.ts
Donation confirm	src/lib/actions/profile.ts → confirmDonation()
Home dashboard	src/app/(app)/home/page.tsx
Profile / history	src/app/(app)/profile/page.tsx, src/components/donor/donation-history.tsx
Auth gates	src/lib/auth.ts
Constants	src/lib/constants.ts (MATCH_RADIUS_KM = 50, DONOR_COOLDOWN_DAYS = 90)
Types	src/lib/types.ts

This document should be updated whenever major product decisions or schema changes are made.